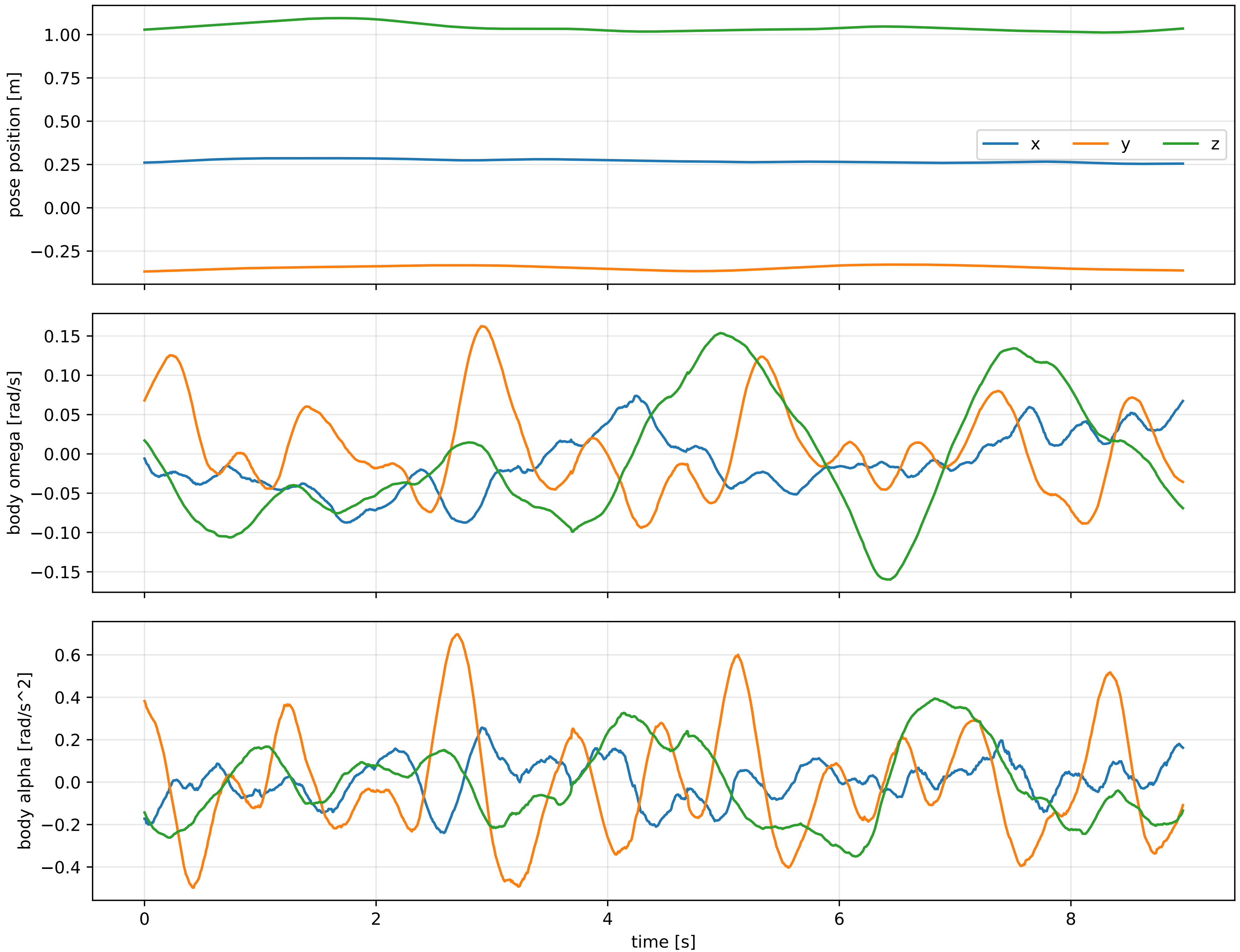
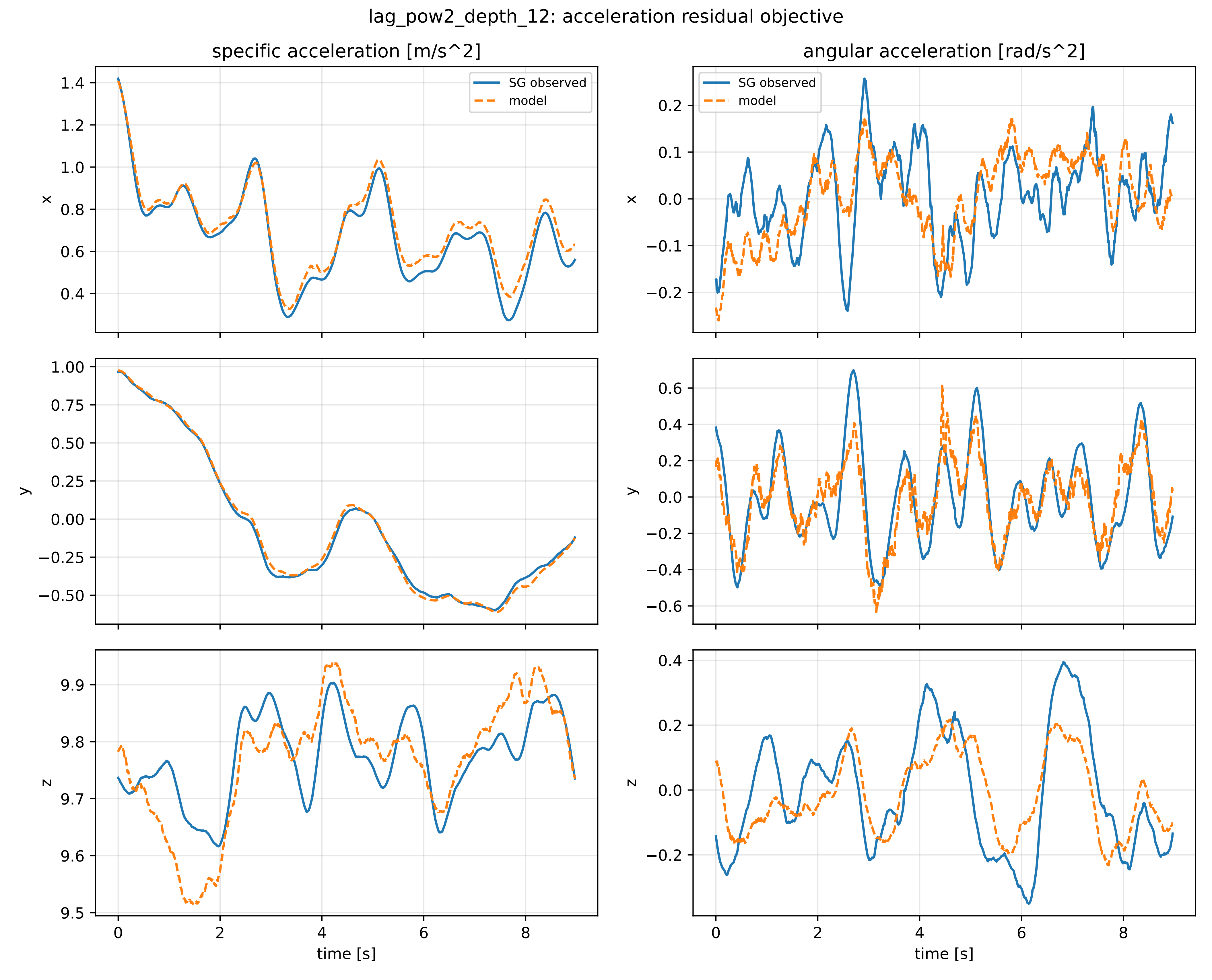
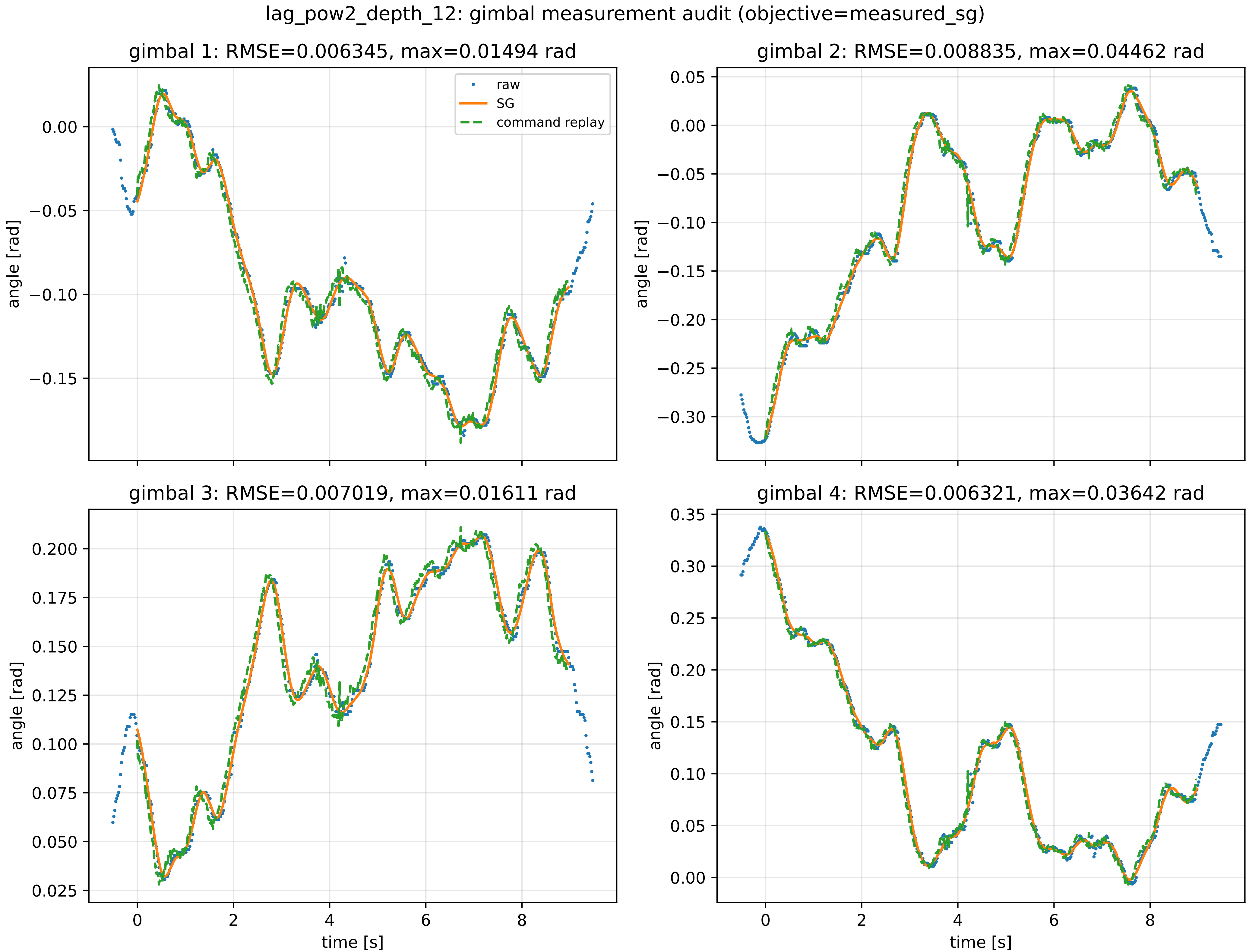
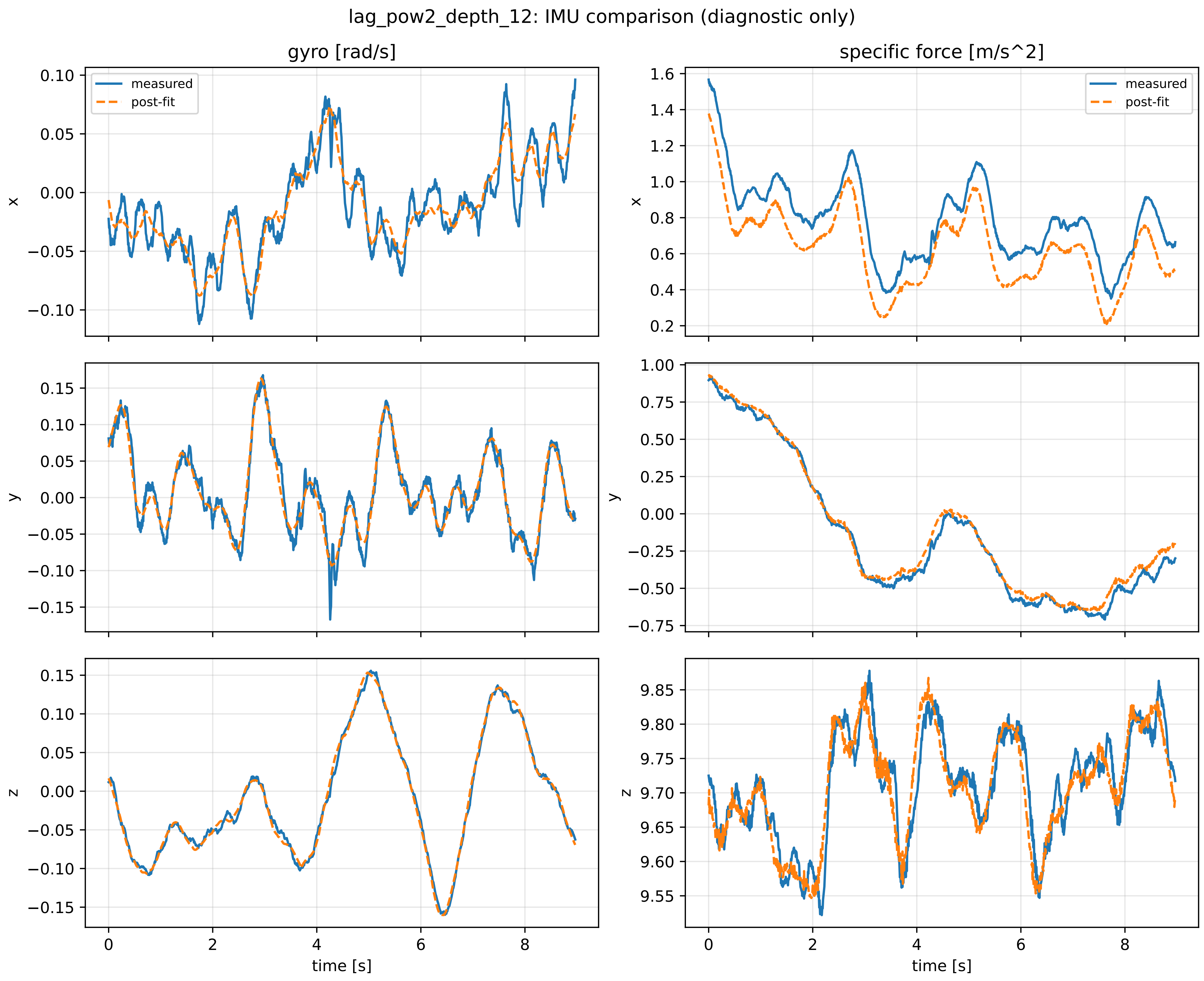
lag_pow2_depth_12: SG trajectory and derived motion



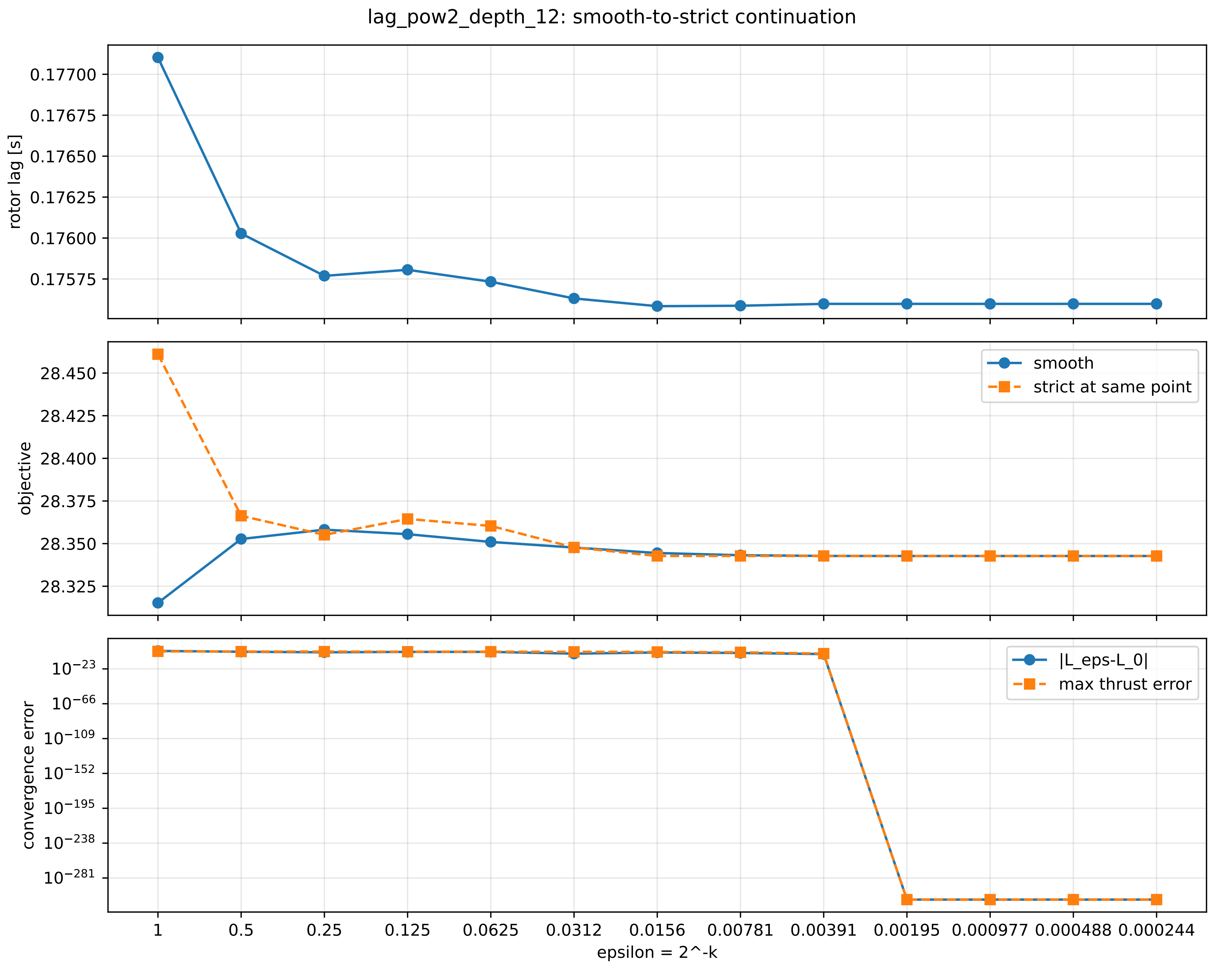


lag_pow2_depth_12: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]

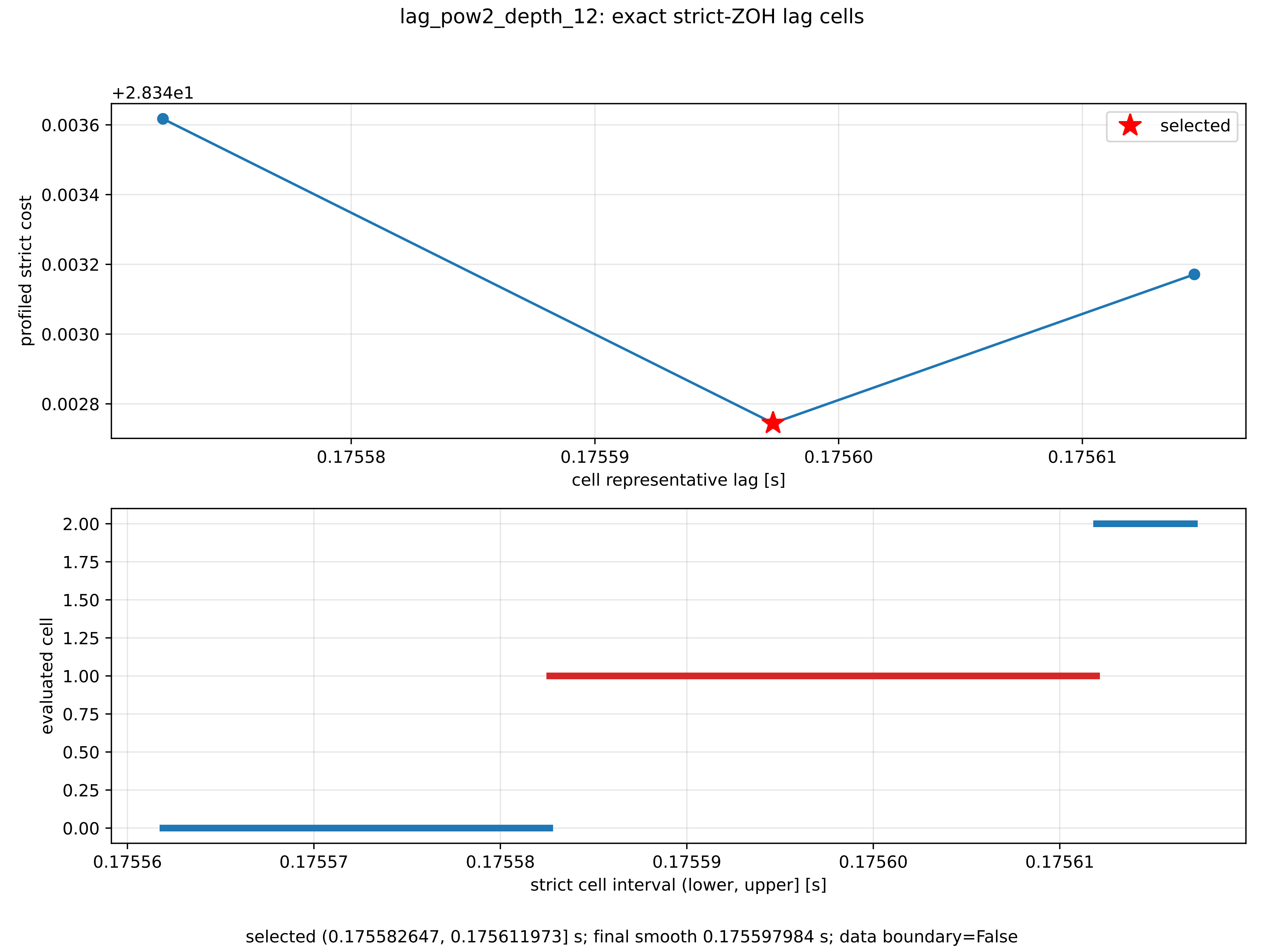

lag_pow2_depth_12: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.006345, max=0.01494 rad
gimbal 2: RMSE=0.008835, max=0.04462 rad
gimbal 3: RMSE=0.007019, max=0.01611 rad
gimbal 4: RMSE=0.006321, max=0.03642 rad
raw
SG
command replay
angle [rad]
time [s]



lag_pow2_depth_12: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]



lag_pow2_depth_12: smooth-to-strict continuation
rotor lag [s]
0.17700
0.17675
0.17650
0.17625
0.17600
0.17575
objective
28.450
28.425
28.400
28.375
28.350
28.325
smooth
strict at same point
convergence error
10^{-23}
10^{-66}
10^{-109}
10^{-152}
10^{-195}
10^{-238}
10^{-281}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
0.000977
0.000488
0.000244
epsilon = 2^-k

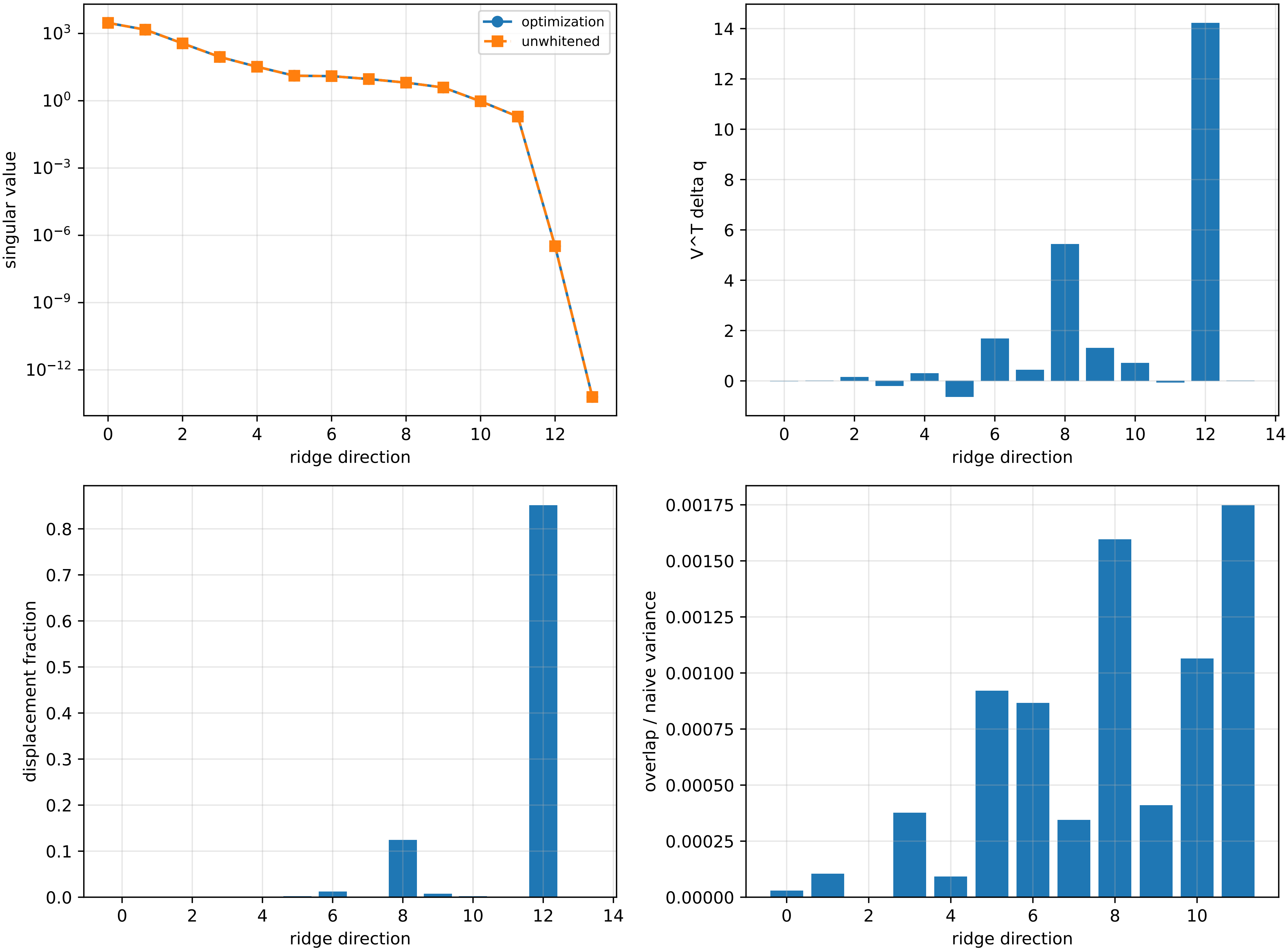


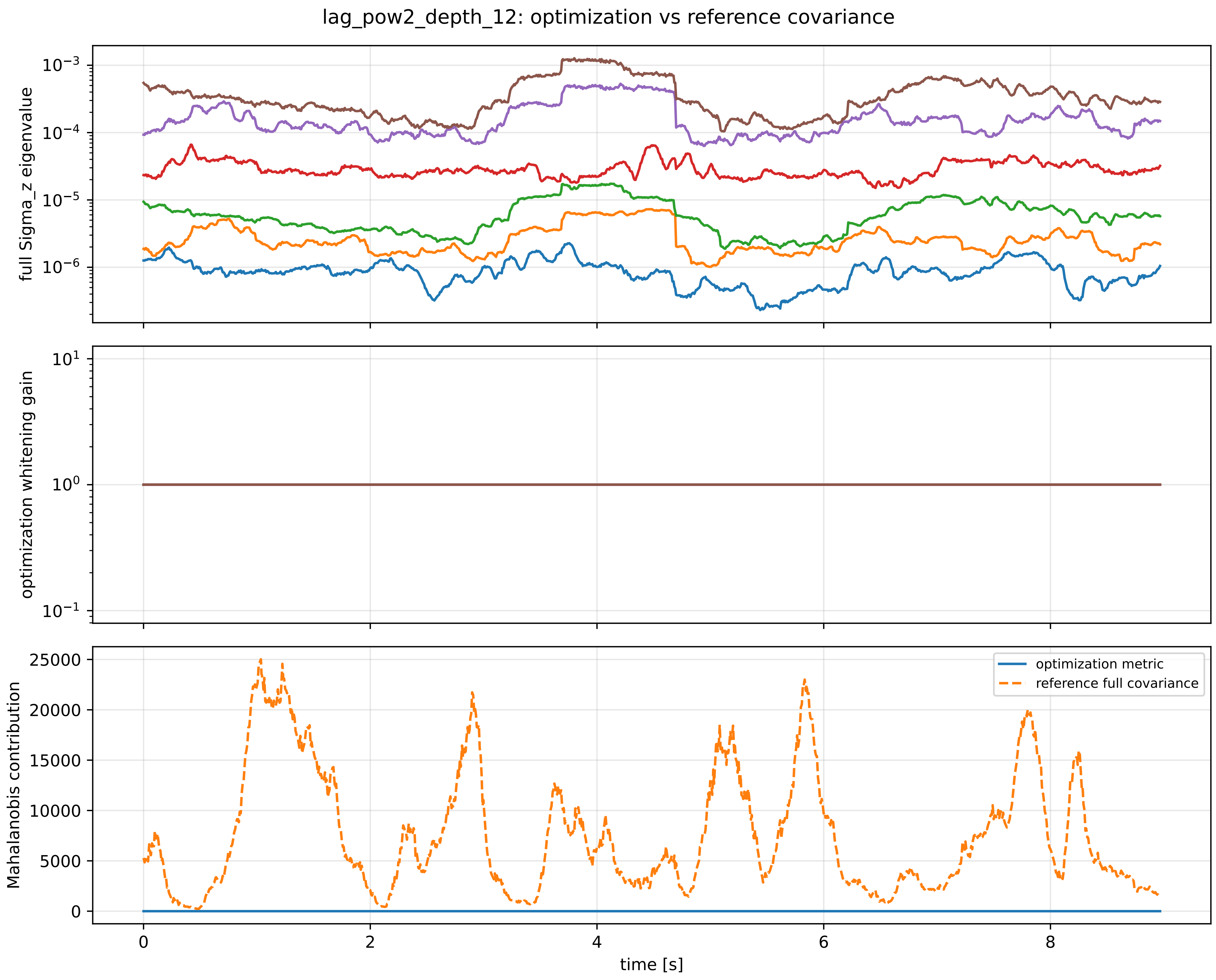
lag_pow2_depth_12: exact strict-ZOH lag cells
+2.834e1
0.0036
0.0034
0.0032
0.0030
0.0028
profiled strict cost
selected
0.17558
0.17559
0.17560
0.17561
cell representative lag [s]
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
evaluated cell
0.17556
0.17557
0.17558
0.17559
0.17560
0.17561
strict cell interval (lower, upper] [s]
selected (0.175582647, 0.175611973] s; final smooth 0.175597984 s; data boundary=False

lag_pow2_depth_12: parameter estimate

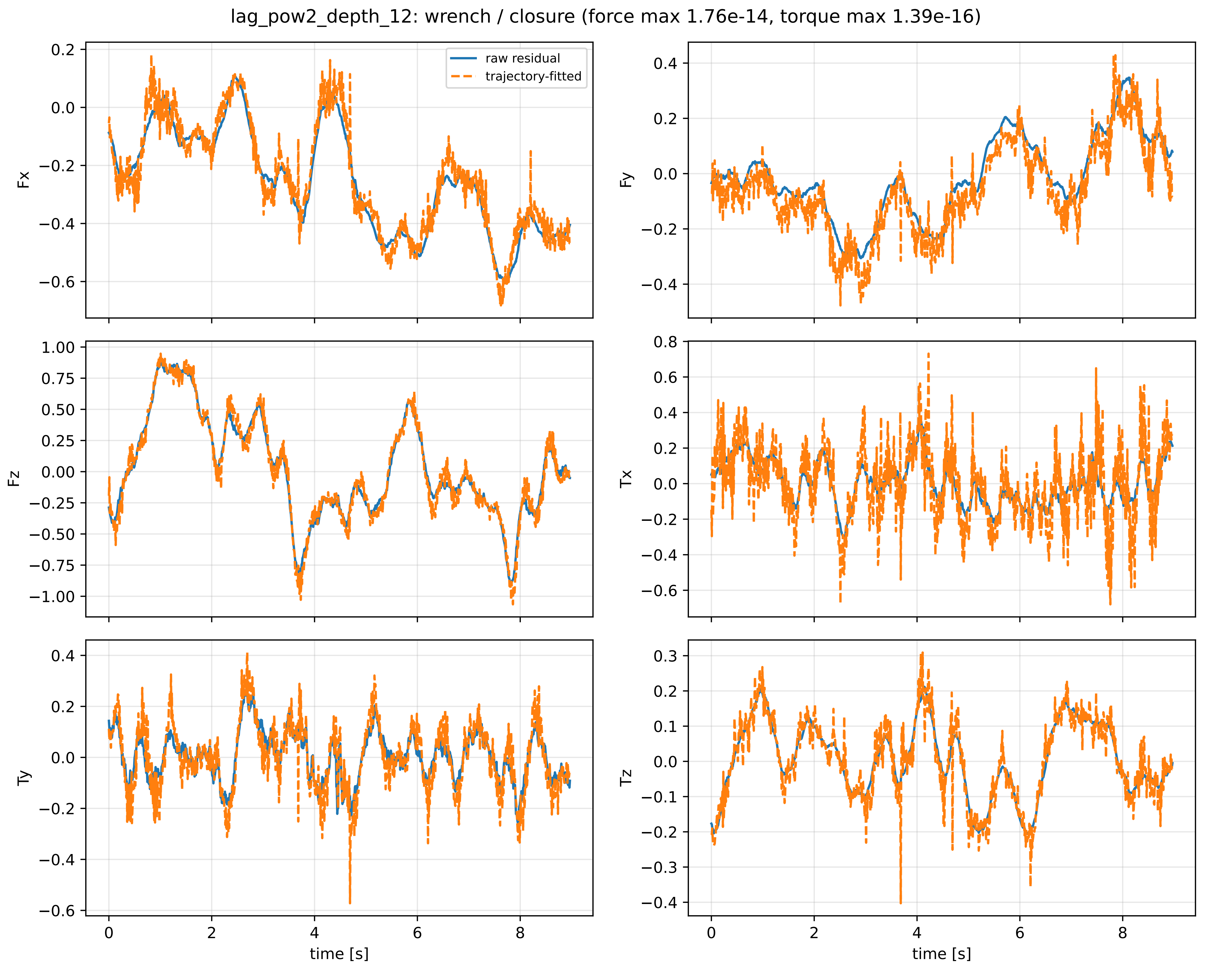
```
Case: lag_pow2_depth_12
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 5.95395078
inertia [kg m^2]:
[[ 1.3808892  -0.03064694  0.10745121]
 [-0.03064694  0.65296997 -0.01821475]
 [ 0.10745121 -0.01821475  0.76743357]]
CoG body [m]: [-0.01982261 -0.05513055 -0.00503048]
force effectiveness: [2.6467129  2.5899967  1.36647135 1.65924684]

scale-free J/m [m^2]:
[[ 0.23192822 -0.00514733  0.01804704]
 [-0.00514733  0.10967003 -0.00305927]
 [ 0.01804704 -0.00305927  0.12889485]]
scale-free f/m: [0.44453053 0.43500472 0.22950666 0.27867997]

rotor lag cell: (0.175582647, 0.175611973] s
representative: 0.17559731 s
```

lag_pow2_depth_12: ridge spectrum (rank 13, nullity 1, ||Jv||=8.26e-14)





lag_pow2_depth_12: optimization vs reference covariance
full Sigma_z eigenvalue
optimization whitening gain
Mahalanobis contribution
optimization metric
reference full covariance
time [s]



lag_pow2_depth_12: wrench / closure (force max 1.76e-14, torque max 1.39e-16)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]